先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 01 – Overview: Learning from Examples

Chizhi Chris ZHANG

zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

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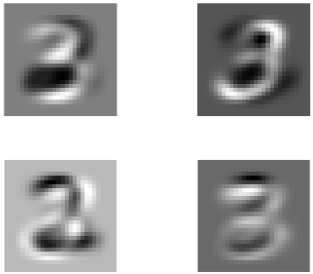
Today: from an observation to a tested prediction

Period	Connected learning task
1 (45 min)	Describe measurements and targets. Use vectors and scales, then fit the six-fruit classifier.
2 (45 min)	Calculate distances, probabilities and losses. Connect boundaries with model assumptions.
3 (45 min)	Build the fish-recognition chain. Evaluate defects, decision costs and a new-writer experiment.

Learning outcome

Specify the data, explain the fitted rule, work through a new prediction and state what independent evidence supports it.

Handwriting varies within a class and overlaps across classes



The four images include different handwritten forms of 2 and 3. A rule based on one stroke can fail when a writer joins or bends that stroke.

- Variation within a class: different writers still mean the same digit.
- Similarity across classes: a poorly written 2 may resemble a 3.
- The goal: predict digits written by people whose examples did not train the model.

Starting question

Which differences should a useful representation preserve, and which should it ignore?

Useful variation and irrelevant variation

Change in a digit	Should its class change?	What a representation should do
Move the same mark slightly	No	Preserve digit identity despite position.
Change the number of loops	Possibly	Retain structure that can distinguish digits.
Increase background brightness	No	Avoid treating illumination as the label.

Invariance has a purpose

Ignoring every difference destroys information. Preserve differences related to the target while reducing sensitivity to changes that should leave the target unchanged.

A testable choice

Compare the same writer under two positions and two lighting conditions. Check whether the representation changes in the intended way.

Measurement error changes the input before learning

$$x_{\text{recorded}} = x_{\text{true}} + b + \epsilon.$$

Term	Meaning	Example
b	Systematic bias	A scale always reads 2 g high.
ϵ	Variable measurement error	Repeated readings differ slightly.
Class label	The target identity	The fruit remains a tomato.

- A constant bias can shift all weights. Random error spreads repeated measurements.
- A classifier cannot distinguish a true property from an instrument error without additional evidence.
- Check calibration and repeated readings before interpreting a class overlap as a model failure.

A pattern is information measured from an object

Physical object	Recorded representation	Target
Handwritten mark	784 grayscale pixel values	Digit class
Spoken utterance	Samples of a waveform	Word sequence
Fruit on a scale	Weight and color measurements	Fruit class

$$\mathbf{x} = (x_1, \ldots, x_{784}) \in \mathbb{R}^{784}, \qquad \widehat{y} = f(\mathbf{x}) \in \{0, 1, \ldots, 9\}.$$

Three distinct things

The object exists in the world. Measurements describe it. The predicted class is the model's answer, which can be wrong.

Learning means improvement on a specified task

Component	Handwriting example
Task T	Predict the digit represented by a new image.
Experience E	Labeled images from training writers.
Performance P	Correct predictions divided by tested images from new writers.

A measurable claim

After learning from E, performance on task T improves according to P. A task without a target or evaluation rule is still underspecified.

Why new examples matter

A table that stores every training image and its answer can have perfect training accuracy while failing on one unfamiliar writer.

Learning changes how a prediction rule is obtained

	Explicit procedure	Learned predictor
Example	Sort numbers ascending	Recognize handwriting
Human design	State the exact comparison rule	Choose data, model family and learning procedure
Role of data	Values on which the rule runs	Examples that determine model parameters
Main check	Does implementation follow the rule?	Does the fitted rule generalize?

The choice depends on the problem

Known exact rules often give reliable software. Learning helps when useful relationships can be estimated from examples but are difficult to enumerate completely.

The course combines representation, statistics and computation

Idea	The question it helps answer
Pattern recognition	Which measurements distinguish categories or structures?
Probability and statistics	How do observations support estimates under uncertainty?
Linear algebra	How can vectors and matrices describe and transform data?
Optimization	How can a computer find parameters with smaller loss?
Machine learning	How can examples improve predictions on new cases?

One connected example

Pixels form a vector. A model assigns a score. A loss measures error. An optimization method adjusts parameters. A held-out set tests the resulting predictor.

Samples, features and labels describe different parts of the data

Sample	Weight / g	Red-pixel fraction	Label y
Fruit 1	40	0.80	Tomato
Fruit 2	50	0.90	Tomato
Fruit 3	110	0.30	Mandarin

- A sample is one fruit in this experiment. Repeated photographs of it are correlated observations.
- Features are measurable input properties with a fixed meaning and order.
- The label is the known target. The prediction $\hat{y}$ is computed from the inputs.

A heavy tomato

An 85 g tomato may receive a mandarin prediction. The measured input, the true label and the predicted label remain separate.

Variable types determine meaningful comparisons

Field	Type	Meaningful operation
Fruit identity	Identifier	Find or join the same fruit.
Weight in grams	Numerical quantity	Compare a difference or compute a mean.
Fruit class	Category	Count classes or check a prediction.
Defect severity	Ordered category	Compare order if the levels have an agreed definition.

Storage does not define meaning

An identifier may use integers. A category may use codes 0 and 1. Neither automatically becomes a physical quantity. Arithmetic is meaningful only when the field definition supports it.

A labeling rule turns judgments into comparable targets

Observation	Ambiguous instruction	Operational instruction
Surface scratch	Looks bad	Visible scratch longer than 2 mm under specified imaging.
Digit mark	Readable	Two annotators assign a digit or record uncertainty.
Fruit damage	Damaged	Apply an agreed visible-surface criterion.

- Labels require a target definition, an observation protocol and a way to handle disagreement.
- Record uncertain cases instead of silently forcing every object into a confident category.
- A model trained on inconsistent targets can learn annotator habits rather than the intended property.

A measurement protocol makes features repeatable

Feature	Procedure to state	Reason
Red-pixel fraction	Camera, illumination and rule for a red pixel	Different settings can change the fraction.
Fruit weight	Calibration, unit and resolution	A stored decimal does not establish measurement precision.
Signal energy	Window duration and sampling rate	Changing the window changes what the value represents.

Repeatable evidence

Another person should be able to obtain the same kind of measurement from the same kind of object. Feature names alone do not specify a procedure.

A dataset is an aligned matrix and target vector

$$X = \begin{bmatrix} 40 & .8 \\ 50 & .9 \\ 60 & .7 \\ 100 & .2 \\ 110 & .3 \\ 120 & .1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} T \\ T \\ T \\ M \\ M \\ M \end{bmatrix}, \qquad X \in \mathbb{R}^{6\times 2}.$$

- $n = 6$ counts observations. $d = 2$ counts input measurements per observation.
- Row i of X and entry i of $\mathbf{y}$ must refer to the same fruit.
- The first column is always grams. The second is always a fraction.
- Additional fruits increase n. Measuring diameter increases d.

A useful check

Shuffle whole rows together with their labels. Shuffling each column independently invents fruits that were never observed.

A feature vector has components with a fixed order

$$\mathbf{x} = \begin{bmatrix} 50 \\ 0.90 \end{bmatrix}, \quad x_1 = 50\ \text{g}, \quad x_2 = 0.90.$$

Expression	What it selects	Interpretation
x_1	First component	Weight of one fruit.
X_{i1}	Row i, column 1	Weight of fruit i.
$X_{:1}$	All rows, first column	A collection of weights.

One object versus a collection

A vector describes one sample. A matrix organizes several samples. The numerical shape is useful only together with the row and column definitions.

Length and direction describe different vector properties

$$\mathbf{a} = (3, 4), \quad \|\mathbf{a}\|_2 = \sqrt{3^2 + 4^2} = 5, \quad \mathbf{u} = \mathbf{a}/5 = (0.6, 0.8).$$

- The Euclidean norm measures distance from the origin when coordinates share an appropriate geometry.
- Dividing by the norm gives a unit vector with the same direction. It removes magnitude information.
- For signal samples, the squared norm can represent energy and the norm represents root-energy magnitude. For a shape descriptor, direction may be the intended information.
- Normalization is a representation choice. Explain what information it discards before using it.

An inner product computes a weighted agreement

$$\mathbf{w} = (1, -2), \quad \mathbf{x} = (4, 1), \quad \mathbf{w}^{\mathsf{T}}\mathbf{x} = 1(4) + (-2)(1) = 2.$$

Input change	Change in the inner product	Why
Increase x_1 by 1	+1	Its coefficient is 1.
Increase x_2 by 1	-2	Its coefficient is -2.
Use the orthogonal vector $(2, 1)$	0	The positive and negative contributions cancel.

Connection to a classifier

Adding a bias gives a score. A later threshold turns the score into a class. The inner product itself returns a number, not a label.

Representations determine which differences a model can use

Representation	Preserves	Can lose or distort
Fruit weight only	Size-related information	Color and shape differences
Weight plus red fraction	Size and color	Illumination changes the fraction
Digit pixels	Local stroke positions	Shifts change many entries
Stroke relations	Connections among parts	Extraction errors in broken strokes

Consistency matters

Training and prediction must use the same feature order, units and measurement procedure. A shape-compatible input can still carry the wrong meaning.

Later connection

Feature engineering and deep learning both address representation, using different ways to construct informative inputs.

Changing units can change a naive distance

$$d^2 = (\Delta\text{weight})^2 + (\Delta\text{red fraction})^2.$$

Comparison	Weight difference	Color difference	Naive squared distance
A	10 g	0.8	100.64
B	20 g	0.1	400.01
A, kilograms	0.01 kg	0.8	0.6401
B, kilograms	0.02 kg	0.1	0.0104

The nearest object can reverse

Only the weight unit changed. A scale-aware representation or a task-specific distance must state how feature differences should trade off.

An ablation asks which measurement helps prediction

Input representation	Validation errors among 20 fruits	Interpretation
Weight only	5	Some heavy tomatoes overlap with mandarins.
Color only	4	Lighting and color similarity remain difficult.
Weight and color	2	The measurements can provide complementary evidence.

- Use the same validation observations and evaluation rule for each comparison.
- Adding a column helps only if it supplies useful information and the model can use it.
- These numbers are a teaching illustration. Select using validation data, then report a separate final test.

The meaning of the target defines the prediction task

Task	Output	From one fruit image
Classification	One category	Tomato or mandarin
Regression	Numerical quantity	Weight in grams
Multilabel prediction	Several applicable labels	Fruit, red and damaged
Ranking	Order among candidates	Which item most resembles this image?

Codes and quantities differ

Coding classes as 0 and 1 does not make them a numerical measurement. A 10 g weight difference has a physical meaning. A difference between two class codes does not.

The same input supports different tasks

Specify the target before selecting an algorithm or a performance measure.

Learning settings differ in the information available

Setting	Training information	Typical goal
Supervised	Inputs and known targets	Predict a class or quantity
Unsupervised	Inputs without target labels	Discover groups or coordinates
Semi-supervised	A few targets, many unlabeled inputs	Improve target prediction
Reinforcement	States, actions and rewards	Improve long-term return

Input format does not define the setting

The same digit images can train a labeled classifier or an unlabeled clustering model. What the learner receives and what it must accomplish determine the setting.

Unsupervised data still contain examples

The absence of labels does not mean the absence of data or assumptions.

A task exercise distinguishes targets from learning signals

Available data	Question	Setting
Fruit images with species labels	Predict species of a new fruit	Supervised classification
Fruit images without species labels	Organize visually similar groups	Unsupervised grouping
Images with measured weights	Predict a numerical weight	Supervised regression
Game states and rewards	Choose actions for later wins	Reinforcement learning

Explain the change

Hiding class labels changes the information available to the learner. It does not remove observations. Changing the target from species to weight changes the task even if the image files stay the same.

Supervised labels and unsupervised groups answer different questions

Supervised fruit task

Known labels: T, T, T, M, M, M. Learn a rule that predicts T or M for new measured fruit. A wrong class can be checked against its label.

Customer	Visits	Spend
A	2	20
B	3	22
C	15	80
D	16	75

Unsupervised customer task

Group similar rows without known group labels. A possible split is $\{A, B\}$ and $\{C, D\}$.

Interpretation requires care

These groups reflect selected measurements and similarity. They do not automatically reveal a true social or business category.

Additional data and interactive feedback supply different learning signals

Example	How learning gets information
Semi-supervised digits	1,000 labeled and 100,000 unlabeled images. Similarity can help if it agrees with digit identity.
Self-supervised text	Hide a token and predict it from context. The data themselves provide training targets.
Reinforcement game	Choose an action, observe a changed state and later reward. A winning game does not label every move as optimal.

The assumption matters

More unlabeled data help only when their structure relates to the task. Delayed rewards require reasoning about action consequences.

Course connection

We return to semi-supervised learning in L12 and to self-supervised pretraining in L18.

Six labeled fruits determine two class centers

Known class	Training weights / g		
Tomato	40	50	60
Mandarin	100	110	120

$$\mu_T = \frac{40 + 50 + 60}{3} = 50 \text{ g}, \qquad \mu_M = \frac{100 + 110 + 120}{3} = 110 \text{ g}.$$

- The model represents each class by its mean weight.
- The learning procedure computes a mean from that class's labeled examples.
- The learned parameters are 50 and 110, rather than the individual training weights.
- We assume that closeness to a class center is informative about its label.

A teaching dataset

These six values illustrate calculation. They do not establish a biological law about fruit weight.

A new training fruit updates a mean with a clear weight

$$\mu_{n+1} = \frac{n\mu_n + x_{n+1}}{n+1} = \mu_n + \frac{x_{n+1} - \mu_n}{n+1}.$$

Tomato calculation

Three training tomatoes have mean 50 g. Add a reliably labeled 70 g tomato: $(3 \cdot 50 + 70)/4 = 55$ g. The new value contributes one quarter to the updated mean.

- The update uses a new training observation and its reliable class label.
- Prediction alone does not justify adding an unlabeled fruit as its own predicted class.
- The formula avoids re-summing old values, but it still performs learning by changing a parameter.

One extreme training value can move a class center

Tomato training weights / g	Mean / g	Median / g
40, 50, 60	50	50
40, 50, 180	90	50

A changed parameter changes prediction

With mandarin mean 110, the mean-center midpoint moves from 80 g to 100 g. A query at 95 g changes from mandarin to tomato.

- First determine whether 180 is a valid fruit weight or a measurement/label error.
- A robust center may reduce sensitivity, but choosing it changes the model and needs validation.
- An unusual valid observation should not disappear solely because it makes the average inconvenient.

Prediction compares a new fruit with fixed learned centers

$$\widehat{y}(x) = \arg\min_{c \in \{T, M\}} |x - \mu_c|.$$

New weight / g	$\lvert x - 50 \rvert$	$\lvert x - 110 \rvert$	Prediction
75	25	35	Tomato
95	45	15	Mandarin
80	30	30	Mandarin by convention

Reading the notation

Evaluate both distances. $\arg\min$ returns the class that gives the smaller distance, not the distance value.

Training and prediction have different jobs

Prediction uses the existing centers. A new input does not automatically change them. At equal distance, our implementation chooses mandarin.

A model family, its parameters and its learning algorithm

```
center_t = sum([40, 50, 60]) / 3
center_m = sum([100, 110, 120]) / 3
def predict(x):
    return "Tomato" if abs(x-center_t) < abs(x-center_m) else "Mandarin"
print(predict(75))  # Tomato
```

Term	In this example
Model family	Nearest class center
Parameters	Two center locations
Learning algorithm	Average each class's training weights
Fitted model	The distance rule with centers 50 and 110

The same model family can use different estimation rules

Fixed model family	Possible estimation rule	What changes
Nearest class center	Mean of each labeled class	Centers reflect all values equally.
Nearest class center	Median in a one-feature task	Centers resist some extreme values.
Linear score	A rule based on classification errors	Weights change in response to wrong signs.
Linear score	An optimizer for a smooth loss	Weights follow a different objective.

Model and algorithm are separate choices

A fitted predictor contains parameter values. An estimation procedure explains how those values were obtained from examples.

The decision boundary follows from equal distances

$$(x-50)^2 = (x-110)^2 \quad \Longrightarrow \quad 120x = 9600 \quad \Longrightarrow \quad x = 80.$$

$$\widehat{y}(x) = \begin{cases} T, & x < 80, \\ M, & x \geq 80. \end{cases}$$

- A boundary separates input regions with different predicted classes.
- In one feature it is a point. In two features it can be a curve or line.
- For two centers under ordinary Euclidean distance, the boundary is their perpendicular bisector.
- The boundary describes a fitted decision rule. Real class distributions may overlap across it.

A quick check

At 79 g the model predicts tomato. At 81 g it predicts mandarin. The true fruit identity need not change at 80 g.

Two-feature centroid prediction is a complete vector calculation

$$\boldsymbol{\mu}_T = (0,0), \quad \boldsymbol{\mu}_M = (4,2), \quad \mathbf{x} = (1,1).$$
$$d_T^2 = (1-0)^2 + (1-0)^2 = 2, \qquad d_M^2 = (1-4)^2 + (1-2)^2 = 10.$$

- Coordinates here are already comparable scaled features in a separate teaching example.
- The tomato center is nearer because $2 < 10$. Squared distances give the same ordering as distances.
- Moving the query to $(3,1)$ gives distances 10 and 2, reversing the prediction.
- Each class center is now a vector. Both components participate together in the decision.

Equal vector distances produce a linear boundary

$$\|\mathbf{x}\|^2 = \|\mathbf{x} - (4,2)\|^2 \quad \Longrightarrow \quad 8x_1 + 4x_2 = 20.$$

Input	Left side	Decision
$(1,1)$	12	Closer to $(0,0)$.
$(2,1)$	20	Equal distance: specify a tie rule.
$(3,1)$	28	Closer to $(4,2)$.

What cancels

The squared input terms cancel when the two distances are subtracted. The remaining weighted sum describes a straight line. This result depends on using the same Euclidean geometry for both centers.

A hard label and a probability express different information

Output	Meaning
Mandarin	A chosen category
$P(Y = M \mid \mathbf{x}) = 0.65$	Estimated probability of mandarin conditional on the measurements
Distance 15 g	Difference between a weight and a center, in physical units

Probability predictions allow uncertainty

Probabilities of all mutually exclusive classes sum to one. A probability of 0.65 leaves 0.35 for the other class in a binary task.

Distance needs an additional model

The centroid calculation alone does not produce calibrated probabilities. A smaller distance cannot be read directly as a confidence percentage.

Later connection: L03

Generative classifiers use probability distributions and Bayes' rule to obtain class probabilities.

A probability table assigns mass to possible outcomes

Outcome	Count among 10 observations	Estimated probability
Tomato	6	0.6
Mandarin	4	0.4
Either class	10	1.0

- Probabilities are nonnegative and exhaustive mutually exclusive outcomes sum to one.
- An observed class frequency estimates a population probability under a sampling interpretation.
- The next individual outcome remains uncertain even when a probability is estimated well.
- Repeated photos of one fruit do not provide the same evidence as ten independent fruits.

Conditional probabilities use a selected denominator

Among 100 fruits	Red	Not red	Total
Tomato	48	12	60
Mandarin	8	32	40
Total	56	44	100

$$P(\text{red} \mid T) = 48/60 = 0.8, \qquad P(T \mid \text{red}) = 48/56 \approx 0.857.$$

The direction matters

The numerator describes the same 48 fruits. Conditioning on tomato selects 60 fruits, while conditioning on red selects 56. Reversing the condition changes the question and the denominator.

Joint and conditional probabilities describe the same counts

$$P(T, \text{red}) = P(\text{red} \mid T)P(T) = 0.8(0.6) = 0.48.$$
$$P(\text{red}) = 0.8(0.6) + 0.2(0.4) = 0.56.$$

- The joint probability describes observing both properties in one sample.
- The total probability adds disjoint contributions from tomatoes and mandarins.
- Independence would require $P(T, \text{red}) = P(T)P(\text{red})$; here $0.48 \neq 0.336$.
- A factorization involving a conditional probability is always valid when the denominator is positive. Replacing it by an unconditional probability requires an independence assumption.

A density height differs from a probability mass

$$X \sim \text{Uniform}[40, 60], \qquad p(x) = 1/20 \text{ per gram.}$$

$$P(45 \le X \le 50) = \int_{45}^{50} \frac{1}{20}\, dx = 5/20 = 0.25.$$

- A continuous density assigns probability through area over a range.
- The density value 0.05 has inverse-gram units. It is not a 5% probability at one exact weight.
- For this continuous model, the probability of one exact value is zero, although intervals around it have positive probability.
- The uniform distribution is an illustration, not a fitted biological model of fruit weights.

Unseen fruits reveal an error that training examples hide

Weight / g	True class	Prediction	0–1 loss
75	T	T	0
95	M	M	0
85	T	M	1
115	M	M	0

$$\text{accuracy} = 3/4 = 75\%, \qquad \bar{\ell} = (0 + 0 + 1 + 0)/4 = 0.25.$$

Generalization

The model must work beyond the exact fruits that determined its parameters. The heavy tomato shows that weight alone can be insufficient.

A useful response to an error

Inspect the failed case. Consider an additional informative measurement or a different model. Testing four cases alone cannot establish a precise population accuracy.

Expectation averages outcomes using their probabilities

$$P(W = 5) = 0.6, \quad P(W = 15) = 0.4, \quad E[W] = 5(0.6) + 15(0.4) = 9 \text{ min}.$$

Quantity	Meaning
Possible outcome	A single order waits 5 or 15 minutes in this toy model.
Expected wait	The probability-weighted average is 9 minutes.
Observed mean	A finite collection can average above or below 9.

Expectation need not be an available outcome

No single order waits exactly 9 in this model. An average describes a collection or distribution rather than a compulsory individual result.

Variance describes spread around an average

$$x = (40, 50, 60), \quad \bar{x} = 50, \quad \frac{1}{3}\sum_i (x_i - 50)^2 = \frac{200}{3}\ \mathrm{g}^2.$$

Summary	Value	Interpretation
Mean	50 g	Center of the three observations.
Mean squared deviation	66.67 g squared	Spread under the divisor-3 convention.
Standard deviation	8.16 g	Square root restores the weight unit.

State the convention

Using divisor $n-1$ instead gives sample variance 100. Software and formulas must agree on the denominator before their outputs are compared.

A loss turns a target-dependent mistake into a number

Classification

$\ell(y, \widehat{y}) = \mathbb{1}\{y \neq \widehat{y}\}$. Every incorrect category has loss 1 under this choice. Our four fruit losses average 0.25.

Regression

$\ell(y, \widehat{y}) = (y - \widehat{y})^2$. A true weight of 90 g predicted as 80 g gives loss $10^2 = 100$ g^2. Predicting 60 g gives $30^2 = 900$ g^2.

$$\widehat{R}(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f(\mathbf{x}_i)).$$

Empirical risk

Average loss on the observed examples provides a criterion for fitting or comparing predictors. The loss must match the question and the consequence of an error.

Average loss can weight different mistakes differently

Outcome among 20 cases	Count	Loss per case	Contribution
Correct	16	0	0
Minor numerical error	3	2	6
Severe error	1	10	10

$$\widehat{R} = (16 \cdot 0 + 3 \cdot 2 + 1 \cdot 10)/20 = 0.8.$$

- The wrong-case fraction is $4/20 = 0.2$, while average cost is 0.8 under the stated loss.
- Both summarize the same outcomes but answer different questions.
- A loss definition should reflect the target and decision consequence rather than be chosen solely to make a score look small.

Data splitting separates learning, choosing and evaluating

Data	What they determine
Training	Parameters, such as class means or network weights
Validation	Choices, such as weight alone versus weight plus color
Test	A performance estimate after the previous choices are fixed

- Split by the unit that the system must generalize to, such as a new writer or a new patient.
- Repeated images of one object can make randomly separated rows too similar.
- Repeated test-set tuning uses test information for model selection.

Handwriting example

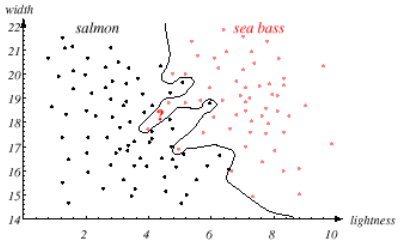
Train and validate using some writers. Test using held-out writers if the goal is to recognize new people's handwriting.

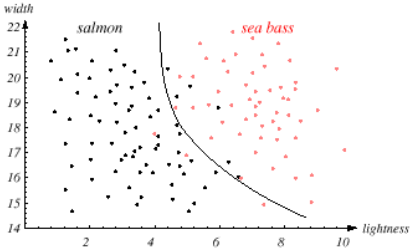
A small test score has limited numerical resolution

Correct / tested	Accuracy	Effect of one changed prediction
3 / 4	75%	25 percentage points.
75 / 100	75%	1 percentage point.
750 / 1000	75%	0.1 percentage point.

- The same percentage can be based on very different amounts of evidence.
- More independent test observations reduce some sampling uncertainty, but cannot repair leakage or an unrepresentative population.
- Report the test count and intended sampling unit together with accuracy.
- A four-fruit example explains calculation rather than establishes precise production performance.

Model complexity can follow the signal or memorize the sample





- Both diagrams show the same salmon and sea-bass measurements: width and lightness.
- The tangled boundary closely follows individual training points. Small measurement changes can cross many bends.
- A smoother boundary tolerates some training errors while describing a more stable separation.
- Compare candidate complexity using new validation cases. A complicated boundary can be useful when enough data support it.

Validation compares complete candidate predictors

Candidate	Training errors / 30	Validation errors / 20	Choice
Simple boundary	4	3	A useful starting point.
Very flexible boundary	0	6	Training improvement did not transfer.
Moderate boundary	2	2	Best among these candidates here.

Selection uses a separate question

Training fits parameters. Validation compares the resulting predictors under an agreed measure. Preserve a final test set after choosing complexity.

Interpret the numbers

Zero training errors can coexist with a larger validation error. The observation supports a comparison on these data, not a universal claim against flexible models.

Historical changes repeatedly expand representation and learning

Development	Main idea	Lasting question
Statistical recognition	Describe class distributions	How should uncertainty affect a decision?
Structural recognition	Describe parts and relations	What structure distinguishes an object?
Perceptrons and kernels	Learn boundaries in chosen representations	Which patterns can the model express?
Deep networks	Learn layered representations	How can many parameters learn useful features?
Pretrained models	Reuse learned representations across tasks	What transfers, and what changes?

The common thread

The field keeps revisiting representation, learning from finite data and performance on new cases. We use the history to connect ideas rather than memorize a list of dates.

Statistical and structural descriptions preserve different evidence

View	Representation	Example reasoning
Statistical	A vector and its class-dependent distribution	Weight and color give evidence for fruit identity even when classes overlap.
Structural	Parts and the relations among them	A digit may contain connected strokes, loops and relative positions.

A representation can combine both

Measured properties of parts can form a vector. A learned system can also model interactions among those parts.

Extraction affects recognition

A broken image stroke may change its inferred connections. A noisy weight measurement may move a fruit across a boundary. The model only receives the representation we construct.

Generative and discriminative approaches describe different quantities

Approach	Quantity modeled	How a decision follows
Generative classifier	$p(\mathbf{x} \mid y)$ and $p(y)$	Compare posterior probabilities using Bayes' rule.
Probabilistic discriminative	$p(y \mid \mathbf{x})$	Choose a class using the directly modeled posterior.
Decision function	Class scores or boundary	Select the largest score or the side of a boundary.

The distinction concerns modeling

A generative classifier describes class-related input distributions. A discriminative method focuses on target prediction. Both can classify the same observations.

Our fruit model

Nearest-center prediction is a distance rule. A probability interpretation requires explicit assumptions about spread and class frequencies.

Nearest neighbors use local examples instead of one class average

Stored training weight / g	Class	Distance from 85 g
60	Tomato	25
100	Mandarin	15
110	Mandarin	25

Nearest-neighbor prediction

Find a close training example and use its label. With more than one neighbor, aggregate the nearby labels. Stored examples directly participate in prediction.

- For 85 g, the nearest listed example is the 100 g mandarin.
- More neighbors smooth the decision, but can ignore a small local class region.
- Distance depends on units and feature scales. A weight difference and a color-fraction difference cannot be compared carelessly.

Later connection

L04 develops distance-based classification and the role of local neighborhoods.

A linear score adds feature contributions

$$s(\mathbf{x}) = w_1 x_1 + w_2 x_2 + b, \qquad \widehat{y} = \mathbb{1}\{s(\mathbf{x}) \geq 0\}.$$

Part	Interpretation
x_1, x_2	Measured input values
w_1, w_2	Contributions per unit of their respective features
b	Offset that moves the decision boundary
$s = 0$	Boundary between the two predicted classes

One calculation

With $w = (1, -2)$, $b = -1$ and $x = (4, 1)$, the score is $4 - 2 - 1 = 1$. This rule predicts class 1.

Scope of the rule

A straight boundary cannot separate every arrangement of points. Changing weights cannot overcome an unsuitable representation.

XOR shows why a different feature can simplify a boundary

x_1	x_2	Class y
0	0	0
0	1	1
1	0	1
1	1	0

Original inputs

No single straight line separates the two class-1 corners from both class-0 corners.

$$z = (x_1 - x_2)^2 = x_1 + x_2 - 2x_1x_2 \quad \text{for binary inputs.}$$

New representation

The four z values are $0, 1, 1, 0$. The threshold rule $\widehat{y} = \mathbb{1}\{z > 0.5\}$ now solves the table exactly.

What changed

The decision is simple in z, but nonlinear in the original features because it includes their interaction x_1x_2. Kernels and neural networks offer broader ways to build such representations.

Interactions describe combinations that single features miss

$$s(x_1, x_2) = ax_1 + bx_2 + cx_1x_2 + d.$$

Inputs	Interaction x_1x_2	What it adds
$(1, 0)$	0	Only the first feature is present.
$(0, 1)$	0	Only the second feature is present.
$(1, 1)$	1	Both features occur together.

- An additive score assigns independent contributions to each feature. An interaction lets one contribution depend on another feature.
- The XOR representation includes such a product. Other tasks may need combinations such as brightness with illumination or dose with duration.
- Additional terms increase flexibility and still require evaluation on new data.

Decision trees predict by following conditional regions

Test	Action
Red fraction > 0.6?	Yes: predict tomato.
Otherwise, weight < 80 g?	Yes: predict tomato. No: predict mandarin.

Walk one example

A fruit with red fraction 0.3 and weight 75 g follows the second test and reaches the tomato leaf. A fruit with the same color fraction and weight 95 g reaches mandarin.

- Each path combines conditions. Each leaf assigns a prediction to the resulting input region.
- Training chooses tests and leaf predictions from examples.
- More depth creates finer regions, which can also memorize small or noisy groups.

Course connection

L05 develops how trees choose splits and control their complexity.

Neural networks learn compositions of transformations

$$\mathbf{h} = \sigma(W_1\mathbf{x} + \mathbf{b}_1), \qquad \mathbf{s} = W_2\mathbf{h} + \mathbf{b}_2.$$

- **x** contains measured inputs. The hidden vector **h** is a learned representation.
- Weight matrices combine values. A nonlinear function σ changes what the whole model can express.
- Output scores **s** support a prediction. A loss compares that prediction with the target.
- Training adjusts many connected parameters using the effect of each on the loss.

From digits to class scores

Different layers can learn useful patterns in the pixels. We still need targets, a loss and a held-out evaluation.

Later lectures

L15 derives backpropagation. L16 specializes representation to images through convolution.

Clustering and dimensionality reduction organize unlabeled inputs

Method	Output	Question to check
Clustering	A group assignment per sample	Does the similarity measure match the intended grouping?
Dimensionality reduction	Fewer coordinates per sample	Which variation or relationships remain?

Groups and coordinates are different

Customers may form groups by visit patterns. A two-dimensional map of image features instead gives positions that summarize a chosen relationship.

A picture needs interpretation

Nearby points on a map may not preserve all original distances. A cluster number is not a verified class label.

Course connection

L10 studies clustering. L11 studies what projections retain, distort or discard.

A discovered group depends on the distance definition

Customer	Visits per month	Monthly spend
A	2	200
B	3	210
C	10	205

- In raw numerical distance, spending differences can dominate visit differences.
- In standardized coordinates, a change of one standard deviation can receive comparable numerical weight across columns.
- Neither choice automatically defines the most meaningful customer grouping.
- Define the purpose of grouping, explain the geometry, and inspect whether groups remain useful under reasonable scale changes.

Ensembles and relational models extend individual predictors

Idea	How it uses information	Main caution
Bagging	Fit models on resampled datasets and aggregate	Identical errors give little gain.
Boosting	Add models to correct a current predictor	Noisy hard cases can receive excessive attention.
Graph-based learning	Use relationships between observed samples	Edges must describe a meaningful relation.

Combination still requires a task

Several models voting on fruit labels remain a classifier. A graph of similar images can support prediction or grouping, depending on its labels and objective.

Course route

L13–L14 develop ensembles. L12 uses sample relationships for semi-supervised learning. All methods return to representation, assumptions and evaluation.

A recognition system connects measurement to a useful decision

Stage	Fruit inspection example	Output for the next stage
Acquisition	Camera records each fruit	Image and sample identity
Preprocessing	Remove background and normalize imaging	Comparable object image
Representation	Measure weight, color or learned features	Feature vector
Training	Fit model using labeled examples	Fixed parameters
Prediction	Apply model to a new fruit	Class or probability
Evaluation	Compare against independent labels	Error pattern and measured performance

Feedback connects the stages

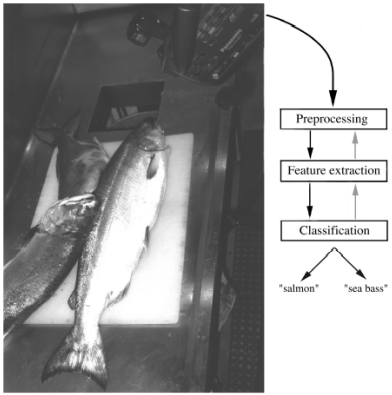
A lighting error may require better acquisition. A missing property may require better representation. Each failed prediction is evidence about the whole chain.

A sampling plan states which population is represented

Sampling choice	What the resulting test can address
One batch, one camera	Performance within that observed batch and imaging condition.
New batches, same camera	Some evidence about batch variation.
Different cameras and times	Evidence about intended measurement changes, if covered adequately.

- Define the future units and conditions before collecting convenient examples.
- Keep repeated views and batch identities so related observations can be split appropriately.
- A large file of convenient images can still omit the difficult conditions that matter.

Fish sorting makes the full recognition chain concrete



- A camera acquires fish images on a conveyor.
- Preprocessing separates each fish from other fish and the background.
- Candidate measurements include length, lightness, width and fin shape.
- A classifier uses those measurements to choose salmon or sea bass.
- Independent labeled fish reveal whether the whole chain works under realistic conditions.

The classifier cannot recover missing evidence

Overlapping objects or unstable illumination can spoil useful measurements before the decision rule runs.

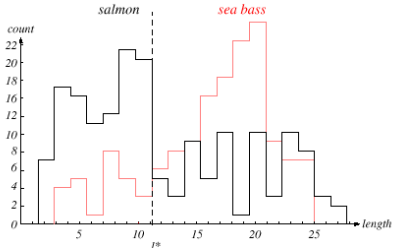
Fish lightness needs an explicit measurement region

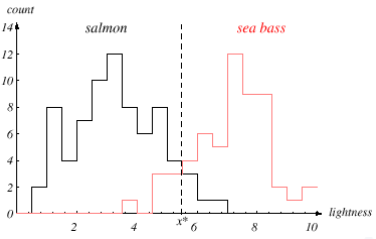
Measurement step	A possible implementation choice	Failure to examine
Object separation	Segment one fish from the background	Background enters the average.
Lightness region	Average a defined body region	A bright fin or reflection dominates.
Illumination control	Calibrate or record the lighting condition	The same species shifts with a new lamp.

A feature is a procedure

The name lightness is only a starting point. Specify the pixels, normalization and conditions that produce its numerical value. Then check whether that value separates the intended classes.

A useful fish feature separates distributions more clearly



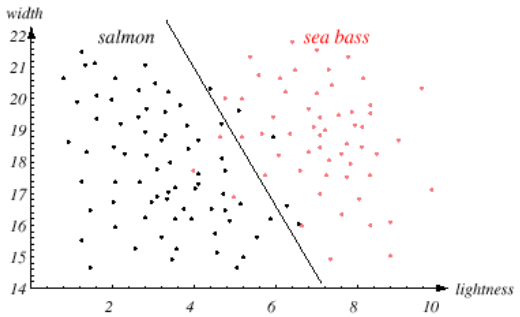


- Black outlines represent salmon. Red outlines represent sea bass. Vertical axes count observed fish.
- Length distributions overlap substantially. Lightness gives a clearer separation in these observations.
- The dashed threshold still leaves errors on both sides. Useful does not mean perfect.
- Select a representation using training and validation evidence, then evaluate it on new fish under the intended lighting conditions.

Connecting the concepts

Feature choice changes the available evidence. A threshold converts that evidence into a prediction.

Two fish measurements make one boundary easier to read



- Horizontal: lightness. Vertical: width. Black and red points represent the two fish classes.
- The boundary combines both measurements. Moving horizontally or vertically can change its score.
- Some observations lie on the unexpected side. A line is a decision rule rather than a statement that all classes are perfectly separate.

Reading guidance

Choose one point from each class near the line. State its coordinates qualitatively and explain which side would determine its prediction.

A feature can supply evidence without uniquely identifying a class

100 labeled fish in a toy region	Salmon	Sea bass	Total
Light region	6	34	40
Dark region	44	16	60
Total	50	50	100

$$P(\text{sea bass} \mid \text{light}) = 34/40 = 0.85.$$

- Lightness is informative here, but six salmon also occupy the light region.
- A hard decision can be reasonable while still making errors.
- Additional measurements may reduce uncertainty, but overlap can remain when the observation lacks discriminating information.

Training-only preprocessing prevents information leakage

$$\mu_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{i \in \text{train}} x_i, \qquad z = \frac{x - \mu_{\text{train}}}{s_{\text{train}}}.$$

- Estimate normalization and missing-value rules using training data.
- Apply the same fitted rules to validation and test data.
- Keep feature meanings, ordering and sample identities aligned.
- Split related samples together when the goal concerns new objects or new people.

A numerical example

If training mean is 50 and training standard deviation is 10, a new measurement of 70 becomes $z = 2$. Its own value does not change the fitted mean.

Leakage

Using test observations to choose transformations or features can make the final test less independent of model development.

A preprocessing exercise keeps the test value out of estimation

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Training observations

Weights are 40, 50 and 60 g. The centering rule uses their mean 50 g. A test weight is 90 g.

Procedure	Centered test value	What it uses
Training mean only	$90 - 50 = 40$ g	A fixed rule estimated before testing.
Pool training and test	$90 - 60 = 30$ g	The test value changes the estimated rule.

Explain the difference

The four pooled values average 60 g. That calculation is arithmetically correct but does not implement the stated training-only procedure. Similar leakage can occur in imputation and feature selection.

A prediction may use only information available at that moment

Proposed task	Available before prediction	Revealed later
Predict waiting time	Queue and order time	Actual wait and pickup time
Inspect a product	Its image and permitted measurements	Final laboratory result
Recommend an item	Past interactions	Whether the user clicks it

Availability is part of the input definition

A field in a historical table may have been recorded after the predicted event. Training on that field teaches a rule that cannot run when needed.

Leakage example

Predicting waiting time from the pickup timestamp gives away the outcome. Queue length measured before ordering is a legitimate candidate input.

A majority baseline exposes what accuracy can hide

True condition	Count among 100 products	Always predict normal
Normal	95	95 correct
Defective	5	0 detected

$$\text{accuracy} = 95/100 = 95\%, \qquad \text{defect recall} = 0/5 = 0.$$

The score answers a limited question

Overall accuracy counts correct products. It does not say whether the rare, important class is found.

- Compare a learned model with a simple baseline on the same cases.
- Inspect performance by class and by relevant conditions.
- State the sample count and class frequencies alongside a reported score.

Our fruit case

A balanced small example and an imbalanced production line can need different evaluation emphasis.

A changing class frequency changes a majority baseline

Test population	Normal	Defective	Always-normal accuracy
Population A	95	5	95%
Population B	60	40	60%

- The prediction rule is identical, but its overall score changes with the class mixture.
- Per-class recall reveals that the rule detects zero defects in both populations.
- Compare models on the same intended population and report the class frequencies.
- A score from one mixture does not automatically describe another production context.

A confusion matrix identifies which kind of error occurred

	Predicted defect	Predicted normal	Total
True defect	TP = 8	FN = 2	10
True normal	FP = 4	TN = 86	90
Total	12	88	100

- True positives are detected defects. False negatives are missed defects.
- False positives are unnecessary alarms on normal products.
- True negatives are normal products correctly accepted.
- Check the row and column definitions before reading the numbers.

An error has a consequence

The two missed defects may reach customers. The four false alarms may trigger unnecessary reinspection. Their costs need not be equal.

False-positive rate uses normal cases as its denominator

$$\mathrm{FPR} = \frac{FP}{FP + TN} = 4/90 \approx 4.44\%, \quad \mathrm{FNR} = \frac{FN}{TP + FN} = 2/10 = 20\%.$$

Measure	Selected population	Question
False-positive rate	All truly normal products	How often is a normal product alarmed?
Precision	All alarmed products	How often is an alarm truly defective?
False-negative rate	All truly defective products	How often does a defect escape?

Similar names do not mean equal fractions

FPR and one minus precision have different denominators. First identify the population, then calculate.

Precision, recall and accuracy use different denominators

$$\text{precision} = \frac{TP}{TP + FP} = \frac{8}{12} = 66.7\%, \quad \text{recall} = \frac{TP}{TP + FN} = \frac{8}{10} = 80\%.$$

$$\text{accuracy} = \frac{TP + TN}{N} = \frac{94}{100} = 94\%.$$

Measure	Verbal question
Precision	Of the products receiving a defect alarm, how many are truly defective?
Recall	Of all truly defective products, how many did the system find?
Accuracy	Of all products, how many received the correct class?

Choosing a measure

A reinspection workload concerns the alarms. A missed-defect requirement concerns the true defects. Both can matter in one system.

F1 combines precision and recall with a harmonic mean

$$F_1 = \frac{2PR}{P+R} = \frac{2TP}{2TP+FP+FN} = \frac{16}{22} \approx 0.727.$$

- Here precision is $8/12$ and recall is $8/10$, using the same confusion matrix.
- A very low value of either precision or recall limits F1. It differs from their ordinary arithmetic average.
- F1 omits true negatives and does not encode the actual cost of a missed defect or a false alarm.
- It is a useful summary for some class-focused comparisons, while individual rates and operational consequences remain necessary.

Two thresholds can have the same accuracy and different behavior

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Policy on 100 products	TP	FN	FP	TN
Stricter	6	4	2	88
More sensitive	8	2	4	86

Both accuracies $= 94\%$, recalls $= 60\%, 80\%$.

- The more sensitive policy exchanges two additional false alarms for two fewer missed defects.
- Equal accuracy hides this change because it counts both error types equally.
- Choose using validation evidence and an explicit requirement or cost. The table illustrates outcomes rather than a guaranteed threshold sweep on unspecified data.

Unequal error costs can favor a lower-accuracy policy

Policy	Missed defects	False alarms	Total errors
A	2	4	6
B	1	8	9

$$C_A = 2(100) + 4(1) = 204, \qquad C_B = 1(100) + 8(1) = 108.$$

Explicit assumptions

Each missed defect costs 100 units and each false alarm costs 1. Correct decisions have zero cost here. Under these values, policy B has lower total cost despite more errors.

- Costs describe the decision consequence, not just label frequency.
- Different applications can assign different values.
- Report the cost assumptions so another person can judge the comparison.

Connection to probability

Class probabilities combined with action costs can determine a decision threshold. L03 develops this reasoning.

An uncertain case can be sent for a second examination

Toy decision costs

A wrong automatic class costs 10. A second examination costs 2. Correct automatic classification costs 0.

Largest class probability	Expected automatic loss	Second-examination loss	Preferred action
0.9	$10(1-0.9)=1$	2	Classify
0.6	$10(1-0.6)=4$	2	Examine again

A reject option adds an action

The goal is a lower expected cost, not a claim that every input must receive a confident automatic answer. State examination capacity and assumptions before using this policy.

A threshold moves the trade-off between missed cases and alarms

Defect score	True label	Alarm at $t = .7$	Alarm at $t = .4$
.9	Defect	Yes	Yes
.6	Defect	No	Yes
.5	Normal	No	Yes
.2	Normal	No	No

Decision rule

Raise an alarm when the score is at least t. Lowering t adds cases to the alarm set. In this example it detects an additional defect and creates an additional false alarm.

- The model's scores can stay fixed while the operating policy changes.
- Choose a threshold using validation evidence and explicit costs or requirements.
- Evaluate the chosen threshold on separate test cases.

Scores may need calibration

A score of 0.6 is a probability only when its definition and calibration support that interpretation.

Changing measurement conditions can change the model input

Change	Effect on observed inputs	What to examine
New camera	Different color or sharpness	Comparable calibration and images
New writers	Different stroke styles	Writer-specific error examples
New product batch	Different shapes or defect frequency	Performance by batch and class

A test has a scope

A good score under one set of conditions describes evidence for those conditions. It does not establish performance everywhere.

A concrete diagnosis

If the same normal fruit receives different red fractions after a lamp replacement, inspect acquisition and preprocessing before adding model complexity.

Feedback must include missed cases

Reviewing only alarms can hide defects that the system silently accepted.

Noise and distribution change lead to different diagnoses

Observed situation	Possible source	Useful next check
Repeated readings fluctuate	Measurement noise	Repeat and calibrate the instrument.
All colors shift after a new lamp	Changed acquisition distribution	Compare the same objects before and after.
Only a new product batch fails	Uncovered object variation	Inspect batch-specific features and labels.

- An error pattern is evidence for a diagnosis, not proof of one cause.
- Changing the classifier alone may leave the original source unresolved.
- Collect comparable examples that distinguish competing explanations.

One photograph can support classification, detection or counting

Task	Output for a traffic photograph	Example error
Classification	Whether the image contains a bus	Say no although a bus appears
Detection	Class and location of each vehicle	Miss a bus or place its box incorrectly
Counting	Number of visible vehicles	Count one overlapping vehicle twice
Tracking	Consistent object identities across frames	Assign one vehicle another's identity

Output structure changes evaluation

A category, bounding boxes, a count and a sequence of identities need different target representations and error measures.

System perspective

For autonomous behavior, perception must connect with localization, planning and control. A correct object category alone does not determine the next action.

Face recognition starts with detection and alignment

Stage	Question	Output
Detection	Where is a face in this image?	A face location.
Alignment	How do corresponding landmarks match?	A normalized face region.
Representation	What describes the identity?	A feature vector.
Matching	Which stored identity is most similar?	An identity score or a decision.

Different errors occur at different stages

A missed face prevents later matching. Incorrect alignment can make the same person appear dissimilar. Identification also needs a rule for people absent from the enrolled database.

Sequences and scientific predictions keep the same basic questions

Application	Input and output	Evaluation question
Speech recognition	Waveform to word sequence	Which words are missing, added or replaced?
Text classification	Text to topic or sentiment	Do new documents receive correct labels?
Recommendation	User context to ranked items	Does the ranking help users in future interactions?
Molecular prediction	Molecular information to structure or property	Does independent evidence support the predicted structure or value?

The common framework

Define observations, representation, target, learning experience and evaluation. Scientific predictions can guide experiments. Independent evidence must still test the predicted structure or property.

Class exercise: design recognition for new handwriting writers

Scenario

Collect ten digits from each of 50 writers. You want to recognize digits from a person whose handwriting has never trained your model.

Decision	Explain your choice
Observation and target	What does one sample represent? What is its label?
Representation	How will size, background and feature order stay consistent?
Data split	Which records must stay together?
Baseline and evaluation	What simple model and what error summary will you use?
Failure analysis	Which examples would show whether stroke style matters?

Discuss before revealing the answer

A writer-based test split addresses the stated goal. A random image split may measure an easier task because related handwriting appears on both sides.

Two handwriting experiments measure different abilities

Design	What can be shared	What the test mainly asks
Random image split	The same writers can appear on both sides	New images with partly familiar writing styles.
Writer split	Writers remain separate	Digits from previously unseen writers.

- Both designs can be useful when they match a stated goal. They answer different generalization questions.
- Keep a record of writer IDs, images per writer and preprocessing choices.
- Compare errors by digit and writing condition so the final score remains interpretable.
- For our stated new-writer task, retain a writer-separated final test.

The lecture connects measurements, learning and evidence

Idea	What you should now be able to explain
Data	A sample is an observation unit. Features describe it. A target states the prediction task.
Model and learning	A chosen function family has parameters estimated from training examples.
Prediction and loss	New inputs produce outputs. A task-dependent loss makes their errors measurable.
Generalization	Separate data test the fitted rule under the intended conditions.
Recognition system	Acquisition and representation influence the evidence available to the classifier.

Bridge to L02

The next lecture turns these ideas into arrays and tables. You will learn to keep rows aligned, handle missing measurements and explain what a calculated average actually represents.